ELEC3705 Fundamentals of Quantum Engineering

Laboratory 7 : Defining and Quantifying Entanglement

Abstract

In this lab we will familiarise ourselves with the notion of a tensor product to extend our Hilbert space to higher dimensions, and use this higher order system to explore the nature of quantum entanglement.

Introduction

When considering multi-particle systems, we typically compose our large system by taking a Kronecker tensor product of the separate Hilbert spaces. Consider two electrons, which both have spin of $\frac{1}{2}$. Therefore, they can each be described by the single qubit Hamiltonian

$$\mathcal{H}_1 = \frac{1}{2}\begin{pmatrix} \epsilon_1 & \Delta_1 \\ \Delta_1 & -\epsilon_1 \end{pmatrix}, \mathcal{H}_2 = \frac{1}{2}\begin{pmatrix} \epsilon_2 & \Delta_2 \\ \Delta_2 & -\epsilon_2 \end{pmatrix} \tag{1}$$

where

$\epsilon_{1,2}$ represent the detuning of electron 1 and 2, respectively.

$\Delta_{1,2}$ represent the coupling of electron 1 and 2, respectively.

If we wish to describe these two separate systems as a larger, single system, we can take the Kronecker tensor product of each of them with the identity matrix.

$$\mathcal{H}_1 \otimes \mathbb{1} + \mathbb{1} \otimes \mathcal{H}_2 = \frac{1}{2}\begin{pmatrix} \epsilon_1 & \Delta_1 \\ \Delta_1 & -\epsilon_1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} \epsilon_2 & \Delta_2 \\ \Delta_2 & -\epsilon_2 \end{pmatrix} \tag{2}$$

$$= \frac{1}{2}\begin{pmatrix} \epsilon_1 & 0 & \Delta_1 & 0 \\ 0 & \epsilon_1 & 0 & \Delta_1 \\ \Delta_1 & 0 & -\epsilon_1 & 0 \\ 0 & \Delta_1 & 0 & -\epsilon_1 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} \epsilon_2 & \Delta_2 & 0 & 0 \\ \Delta_2 & -\epsilon_2 & 0 & 0 \\ 0 & 0 & \epsilon_2 & \Delta_2 \\ 0 & 0 & \Delta_2 & -\epsilon_2 \end{pmatrix} \tag{3}$$

$$= \frac{1}{2}\begin{pmatrix} \epsilon_1 + \epsilon_2 & \Delta_2 & \Delta_1 & 0 \\ \Delta_2 & \epsilon_1 - \epsilon_2 & 0 & \Delta_1 \\ \Delta_1 & 0 & -\epsilon_1 + \epsilon_2 & \Delta_2 \\ 0 & \Delta_1 & \Delta_2 & -\epsilon_1 - \epsilon_2 \end{pmatrix} \tag{4}$$

To be clear, here we are mathematically combining two separate systems into 1 ensemble, but there is no interaction between the systems! Note how the dimension of the result is always the product of the dimensions of the components,

$$\dim(\mathcal{H}_1 + \mathcal{H}_2) = 4 \times 4 = (\dim(\mathcal{H}_1) \cdot \dim(\mathcal{H}_2)) \times (\dim(\mathcal{H}_1) \cdot \dim(\mathcal{H}_2)) \tag{5}$$

Also note that our matrices were square, which is true for matrices in a Hilbert space but not true in general. We can similarly construct state vectors in our new Hilbert space, which we will denote $\mathbb{H}^4$, by taking the tensor product of the basis vectors of our smaller Hilbert spaces, which both happen to be $\mathbb{H}^2$.

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{6}$$

$$|\uparrow\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \qquad |\uparrow\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \qquad |\downarrow\uparrow\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \qquad |\downarrow\downarrow\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \tag{7}$$

It is important to keep in mind that the ordering must always be preserved, and so the first and second arrows correspond to the first and second electron spin, respectively. Naturally, we can also apply the tensor product to superposition states of the component spins, which will create superpositions in $\mathbb{H}^4$. One might expect that all valid vectors in $\mathbb{H}^4$ can be constructed in this way, but consider the following example.

$$\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \otimes \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} a_1 b_1 \\ a_1 b_2 \\ a_2 b_1 \\ a_2 b_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} \tag{8}$$

Clearly, none of a_1, a_2, b_1 or b_2 can be zero for the second and third terms to be non-zero. However, taking the first term $a_1 b_1 = 0$, we require that either a_1 or b_1 be zero, which is a contradiction. Therefore it is not possible to deconstruct our $\mathbb{H}^4$ vector into two $\mathbb{H}^2$ vectors. This brings us to the concept of entanglement.

Quantum Entanglement

Quantum entanglement is the phenomenon of having a quantum system composed of discrete elements which can only be described as a whole. To illicit this point, we can calculate the spin value of various states. First, let's define our spin observables

$$\sigma_{z\mathbb{1}} = \sigma_z \otimes \mathbb{1} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \tag{9}$$

$$\sigma_{\mathbb{1}z} = \mathbb{1} \otimes \sigma_z = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \tag{10}$$

Where $\mathbb{1}$ and σ_z are the identity matrix and Pauli z matrix of $\mathbb{H}^2$, respectively. Similar constructions can be made for the x and y Pauli matrices. Note that these matrices each act on a single spin.

We can define a 'total spin operator' for each axis $\sigma_{a,tot} = \sigma_{a\mathbb{1}} + \sigma_{\mathbb{1}a}$, where a is x, y or z. We also define $\sigma^2 = \sigma^2_{x,tot} + \sigma^2_{y,tot} + \sigma^2_{z,tot}$ which we consider to be the squared magnitude of the total spin.

Question 1

Calculate the expectation values $\langle\sigma_{z,tot}\rangle_\psi, \langle\sigma_{x,tot}\rangle_\psi, \langle\sigma_{y,tot}\rangle_\psi$ and $\langle\sigma^2\rangle_\psi$ for the following state vectors:

a) $|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\uparrow\rangle + |\uparrow\downarrow\rangle))$

b) $|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle))$

Describe the meaning of the results.

Density Matrices

A density matrix is an alternative representation to our state vector, which can contain further information of state purity and entanglement. We define a density matrix of a state vector $|\psi\rangle$ to be

$$\rho = |\psi\rangle\langle\psi| \tag{11}$$

Remember that each state vector $|\psi\rangle$ has a corresponding density matrix ρ, but the reverse is not always true. As with state vectors, we can extend our density matrices into higher Hilbert spaces by simply applying the Tensor product.

$$\rho^C = \rho^A \otimes \rho^B \tag{12}$$

This can be 'undone' by applying the formalism of a partial trace.

$$\rho^B = \mathrm{Tr}_A(\rho^C) \tag{13}$$

Note that the ordering of your tensor product must be kept in mind when applying a partial trace. The tensor product is non-commutative.

For the purposes of this lab, we can use the following example

$$\rho^C = \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix} \tag{14}$$

$$\rho^A = \mathrm{Tr}_B(\rho^C) \tag{15}$$

$$= \begin{pmatrix} a+f & c+h \\ i+n & k+p \end{pmatrix} \tag{16}$$

$$\rho^B = \mathrm{Tr}_A(\rho^C) \tag{17}$$

$$= \begin{pmatrix} a+k & b+l \\ e+o & f+p \end{pmatrix} \tag{18}$$

Question 2

Verify these formulations of ρ^A and ρ^B.

Hint: What is the trace of a density matrix equal to?

Entanglement Entropy

We define the entropy of a quantum state by extending the concept of Shannon's entropy in information theory.

$$S = -\sum_i p_i \log(p_i) = -\operatorname{Tr}(\rho \log(\rho)) \tag{19}$$

where

$$p_i \text{ are the diagonal elements of the density matrix } \rho$$

For our spin system in $\mathbb{H}^4$ we can define the 'entanglement entropy' using partial traces to extract the 'component density matrix' of interest.

$$S = -\operatorname{Tr}(\operatorname{Tr}_A(\rho) \log(\operatorname{Tr}_A(\rho))) \tag{20}$$

Here A represents the component space you wish to extricate from your composite density matrix ρ.

Question 3

Calculate the entanglement entropy (w.r.t. the first electron spin, i.e. use $\operatorname{Tr}_B(\rho)$) for the following states by first finding their respective density matrices

a) $|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\uparrow\rangle + |\uparrow\downarrow\rangle)$

b) $|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$

c) $|\psi\rangle = \frac{1}{\sqrt{3}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle + |\downarrow\downarrow\rangle)$

Task

Consider the system Hamiltonian below

$$\mathcal{H} = \frac{\omega_H}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \tag{21}$$

where

$$\omega_H = 2\pi \cdot 10\,\mathrm{MHz} \tag{22}$$

Given the following starting states, perform a time evolution over a full period and calculate the expectation values for the total spin operators and the total spin squared operator, as well as the entropy of entanglement as a function of time.

a) $|\psi\rangle = |\downarrow\downarrow\rangle$

b) $|\psi\rangle = \frac{1}{\sqrt{2}}(|\downarrow\downarrow\rangle + |\uparrow\uparrow\rangle)$